Algorithm 1: AGNES

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input  : D = {x_i}_{i=1}^m, d, k
output: C = {C_1, C_2, ..., C_k}
// 初始化单样本聚类簇
for j = 1, 2, ···, m do
    C = {x_j}
end
// 初始化聚类簇距离矩阵
for i = 1, 2, ···, m do
    for j = 1, 2, ···, m do
        // 这里的d通常使用 d_min，d_max，d_avg
        M(i, j) = d(C_i, C_j)
        M(j, i) = M(i, j)
    end
end
// 设置当前聚类簇的个数 q
q = m
while q > k do
    找出距离最近的两个聚类簇 C_{i*} 和 C_{j*}
    // 合并 C_{i*} 和 C_{j*}
    C_{i*} = C_{i*} ∪ C_{j*}
    for j = j* + 1, j* + 2, ···, q do
        C_j = C_{j-1}
    end
    // 删除距离矩阵 M 的第 j* 行与第 i*列
    for j = 1, 2, ···, q − 1 do
        M(i*, j) = d(C_{i*}, C_j)
        M(j, i*) = M(i*, j)
    end
    q = q − 1
end
return 簇划分C = {C_1, C_2, ···, C_k}
```